Supplementary Information for "Benefit of Inconvenience for Resilient AI-Assisted Plastic Recycling"

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This document contains the complete model specifications, the derivation of parameters, scenario analysis and implementation, model testing results, and comprehensive sensitivity and uncertainty analysis results that support our paper. The model and analysis scripts are available at https://github.com/Alfarisi006/Benefit-of-Inconvenience-for-Resilient-AI-Assisted-Plastic-Recycling.

Table S1. System dynamics (SD) Model equation

Type	Variable	Equation	Unit
Stock	AI Model Accuracy Adv	INTEG (Acc gain Adv-Acc decay Adv, 0.8)	Dmnl
Stock	Audit Intensity Adv	INTEG (Audit adjusment, Min Audit)	Dmnl
Stock	Contamination stock Adv	INTEG (Contamination generation Adv-Contamination removal Adv, 50)	t
Stock	Cumulative emission Adv	INTEG (Total Emission Adv, 0)	t CO_2-eq
Stock	Fuben eki	INTEG (Commitment-Decay, 0)	Dmnl
Stock	Operator Skill Adv	INTEG (Skill gain Adv-Skill decay Adv, 0.8)	Dmnl
Stock (SMOOTH)	Performance expected deterioration	SMOOTH (Performance deterioration , Expected adjustment time)	Dmnl
Stock	Trust AI Adv	INTEG (Trust gain Adv-Trust decay Adv, 0.7)	Dmnl
Auxiliary	Acc decay Adv	AI Model Accuracy Adv*Concept drift rate Adv	1/Month
Auxiliary	Acc gain Adv	ACC learn rate Adv*Audit Intensity Adv*(1-AI Model Accuracy Adv)	1/Month
Auxiliary	Audit adjusment	(Audit Target - Audit Intensity Adv) / Audit adjust time	Dmnl/Month
Auxiliary	Audit Target	Min Audit+(Max Audit-Min Audit)*MIN (1 , Relative performance gap+ 0.6 *Fuben eki)	Dmnl
Auxiliary	Base Error Adv	0.18+0.05*Shock Adv	Dmnl
Auxiliary	Commitment	Fuben eki switch*Learning rate*Relative performance gap*(1-Fuben eki)	1/Month
Auxiliary	Concept drift rate Adv	Concept drift base Adv+Drift extra Adv	1/Month
Auxiliary	Contamination generation Adv	Plastic Input Adv*Sorting Error rate Adv	t/Month
Auxiliary	Contamination removal Adv	MIN (Contamination stock Adv / Rework time , Rework Capacity Adv)	t/Month
Auxiliary	Decay	Forgetting rate*(1 - MIN (1 , Relative performance gap))*Fuben eki	1/Month
Auxiliary	Drift extra Adv	Shock Adv*Drift shock magnitude Adv	1/Month
Auxiliary	Human intervention burden	Audit Intensity Adv*Intervention burden factor	Dmnl
Auxiliary	Performance deterioration	0.5*MIN (1 , Sorting Error rate Adv/Error ref) + 0.5*MIN (1 , Contamination stock Adv/Contam ref)	Dmnl
Auxiliary	Quality Index Adv	1/(1+Quality Sensitivity Adv*Contamination stock Adv/Plastic Input Adv)	Dmnl
Auxiliary	Recycled output Adv	Demand Total Adv*Recycled share Adv	t/Month
Auxiliary	Recycled share Adv	Base recycled share Adv*Quality Index Adv	Dmnl
Auxiliary	Relative performance gap	MAX (0 , (Performance deterioration- Performance expected deterioration) / Performance expected deterioration)	Dmnl
Auxiliary	Shock Adv	STEP(1,30)-STEP(1,36)	Dmnl
Auxiliary	Skill decay Adv	Forget rate Adv*(1-Audit Intensity Adv)*Operator Skill Adv	1/Month
Auxiliary	Skill gain Adv	Learn rate Adv*Audit Intensity Adv*(1-Operator Skill Adv)	1/Month
Auxiliary	Sorting Error rate Adv	Base Error Adv*(1-Human Effect Adv*Operator Skill Adv)*(1-Ai Effect Adv*AI Model Accuracy Adv)*(1+Automation Bias Adv*Trust AI Adv*(1-Verification Effort Adv))	Dmnl
Auxiliary	Total Emission Adv	Virgin demand Adv*EF Resin and Forming Adv+Recycled output Adv*EF Recycled Adv+Contamination removal Adv*EF Rework Adv	tCO_2-eq/Month
Auxiliary	Trust decay Adv	Shock Adv*Shock Sensitivity Adv*Trust AI Adv	1/Month
Auxiliary	Trust gain Adv	Trust speed Adv*(AI Model Accuracy Adv-Trust AI Adv)	1/Month
Auxiliary	Verification Effort Adv	MIN (Max Verification Adv , Audit Intensity Adv * (1 - Trust AI Adv))	Dmnl
Auxiliary	Virgin demand Adv	Demand Total Adv*(1-Recycled share Adv)	t/Month

Simulation control: INITIAL TIME = 0, FINAL TIME = 100 months, TIME STEP = 1 month (Euler), SAVEPER = TIME STEP.

Table S2. Parameter values, units, ranges used in the sensitivity analysis and provenance

Parameter	Value	Unit	Sensitivity range	Provenance
Plastic Input Adv	1,000	t/Month	–	Case-scale throughput (stylized)
Demand Total Adv	900	t/Month	–	Case-scale material demand (stylized)
Rework Capacity Adv	120	t/Month	95–165 (sweep)	Capacity scenario; varied in Table S7
Rework time	1	Month	–	Model assumption
Base recycled share Adv	0.6	Dmnl	–	Behavioural assumption (stylized)
Quality Sensitivity Adv	3	1/Month †	2.25–3.75	Behavioural assumption (stylized)
EF Resin and Forming Adv	2.46	t CO_2-eq/t	–	Author's estimate (Average value)
EF Recycled Adv	0.80	t CO_2-eq/t	–	Author's estimate (Average value)
EF Rework Adv	0.74	t CO_2-eq/t	–	Author's estimate (Average value)
Learn rate Adv (operator)	0.6	1/Month	0.45–0.75	Behavioural assumption (stylized)
Forget rate Adv (operator)	0.1	1/Month	0.075–0.125	Behavioural assumption (stylized)
ACC learn rate Adv	0.4	1/Month	0.30–0.50	Behavioural assumption (stylized)
Concept drift base Adv	0.05	1/Month	0.0375–0.0625	Behavioural assumption (stylized)
Drift shock magnitude Adv	0.1	1/Month	0.075–0.125	Disruption design
Trust speed Adv	0.5	1/Month	0.375–0.625	Behavioural assumption (stylized)
Shock Sensitivity Adv	0.8	1/Month †	0.60–1.00	Disruption design
Automation Bias Adv	0.25	Dmnl	0.1875–0.3125	Behavioural assumption (stylized)
Human Effect Adv	0.6	Dmnl	0.45–0.75	Behavioural assumption (stylized)
Ai Effect Adv	0.7	Dmnl	0.525–0.875	Behavioural assumption (stylized)
Max Verification Adv	0.5	Dmnl	–	Model constant
Min Audit	0.05	Dmnl	–	Intervention bound (scenario assumption)
Max Audit	0.30	Dmnl	0.225–0.375	Intervention bound (scenario assumption)
Audit adjust time	2	Month	1.5–2.5	Response-delay assumption
Learning rate (fuben-eki commitment)	0.20	1/Month	0.15–0.25	Behavioural assumption (stylized)
Forgetting rate (fuben-eki)	0.05	1/Month	0.0375–0.0625	Behavioural assumption (stylized)
Expected adjustment time	6	Month	4.5–7.5	Expectation-delay assumption
Error ref	0.12	Dmnl	0.09–0.15	Normalization reference (= capacity-equivalent error 120/1,000)
Contam ref	120	t	90–150	Normalization reference (= one month of rework capacity)
Intervention burden factor	0.2	Dmnl	–	Indicator scaling (relative results unaffected)
Fuben-eki weight in Audit Target	0.6	Dmnl	–	Embedded constant (Audit Target equation)
Base error / disruption increment	0.18 / 0.05	Dmnl	–	Embedded constants (Base Error Adv equation)
Initial operator skill	0.80	Dmnl	0.40 (robustness)	Commissioning state (newly trained operator)
Initial AI model accuracy	0.80	Dmnl	0.70 (robustness)	Commissioning state (newly deployed model)
Initial trust in AI	0.70	Dmnl	0.60 (robustness)	Commissioning state
Initial contamination stock	50	t	–	Model assumption
Initial audit intensity / fuben-eki state	Min Audit / 0	Dmnl	–	Model assumption
Fuben eki switch	0 / 1	Dmnl	–	Scenario switch (1 in S1 and S3)

Table S3. Scenario specification

Scenario	Fuben eki switch	Shock Adv	Purpose
S0 Conventional, normal	0	0	Baseline automated operation
S1 Fuben-eki, normal	1	0	Routine effect of adaptive involvement
S2 Conventional, disrupted	0	STEP(1,30) − STEP(1,36)	Vulnerability to disruption
S3 Fuben-eki, disrupted	1	STEP(1,30) − STEP(1,36)	Resilience effect of adaptive involvement

The disruption (months 30–36) raises the base sorting error from 0.18 to 0.23, adds 0.10 month^{-1} to AI concept drift and activates trust decay. The design separates the routine effect of fuben-eki (S0 versus S1) from its resilience effect (S2 versus S3); the resilience effect is quantified as the difference-in-differences (DiD) (S3 - S1) - (S2 - S0). The model is a theory-driven, stylized representation; no facility data were available for calibration, so behaviour reproduction against observed data is not reported. Instead, structure and behaviour are assessed through the tests in Table S4.

Table S4. Model testing result

Test	Result	Assessment
Dimensional consistency	All variables carry units. Two unit labels are corrected (no numerical change): Quality Sensitivity Adv and Shock Sensitivity Adv take 1/Month so that C/P (Month) and the trust-decay flow (1/Month) are dimensionally consistent	Pass after unit correction
Scenario consistency	Disrupted and normal runs identical before month 30 (S0 = S2; S1 = S3) at all saved time steps	Pass
Bounds of normalized states (baseline)	Skill 0.24–0.80; AI accuracy 0.24–0.80; trust 0.08–0.76; fuben-eki state 0–0.33; audit 0.05–0.15	Pass
Extreme condition: plastic input → 0	Contamination stock drains to 0; emissions nearly equal across scenarios (156.4 vs 155.4 kt CO_2-eq)	Pass
Extreme condition: rework capacity = 0	No removal; contamination accumulates in all scenarios (12,059 t in S2 at month 100); recycled share → 0.02	Pass
Extreme condition: human and AI effect on error = 0	Error equals the unmitigated base error (maximum 0.251); fuben-eki has no emission benefit	Pass
Extreme condition: maximum audit = minimum audit	No headroom for adaptive audit; S1 = S0 and S3 = S2 exactly	Pass
Extreme condition: concept drift = 0	AI accuracy rises to 0.98; no backlog forms in any scenario	Pass
Extreme condition: fuben-eki commitment rate × 10	Fuben-eki state bounded ($0.93 \leq 1$); stronger benefit (S3 stock 100 t)	Pass
Extreme condition: base recycled share = 0	Material link removed; emissions nearly identical across scenarios (229.6 vs 229.2 kt CO_2-eq)	Pass
Extreme condition: trust shock sensitivity = 1.0 (upper MC bound)	Trust falls marginally below zero (−0.006) with DT = 1 month; non-negative with DT ≤ 0.5 (minimum 0.087). Euler artefact affecting 63 of 500 Monte Carlo samples	Minor (integration artefact)
Integration error (DT 1 → 0.5 → 0.25 → 0.125)	Resilience benefit S2 − S3: 21.27 / 21.14 / 21.06 / 21.04 kt CO_2-eq; S2 stock at month 100: 938 / 927 / 921 / 919 t; change 1 → 0.125 ≤ 2.0%	Pass (DT = 1 retained)
Implementation check (Vensim vs PySD)	Vensim datasets of all four scenarios reproduced by PySD with maximum relative difference $< 1 \times 10^{-6}$	Pass
Initial-condition robustness (0.4 / 0.7 / 0.6)	Resilience benefit 20.2 kt CO_2-eq (10.9%); DiD −7.4 kt CO_2-eq and −293 t (baseline: 21.3 kt, 11.7%, −9.9 kt, −306 t)	Pass (qualitatively robust)
Long-horizon behaviour (300 months, DT = 0.25)	All scenarios converge to error ≈ 0.132; fuben-eki state decays to ≈ 0 as expectations adapt (eroding goals)	As discussed in Section 5

Table S5. One-at-a-time sensitivity result (±25%; values at month 100)

Parameter	Low / high value	Resilience benefit S2 − S3 (kt CO_2-eq)	DiD emissions (kt CO_2-eq)	Extra burden S3 vs S2 (%)	S3 contamination stock (t)
AI effect on error	0.525 / 0.875	11.3 / 5.4	2.9 / −2.3	36 / 44	864 / 110
Human effect on error	0.45 / 0.75	14.2 / 11.1	0.1 / −6.0	38 / 44	604 / 114
Operator forgetting rate	0.075 / 0.125	13.9 / 21.8	−8.6 / −7.3	42 / 41	116 / 306
Base concept drift	0.0375 / 0.0625	15.8 / 21.7	−11.2 / −6.7	41 / 42	117 / 299
Operator learning rate	0.45 / 0.75	21.3 / 17.3	−6.8 / −10.2	41 / 42	338 / 118
AI learning rate	0.3 / 0.5	21.0 / 18.0	−6.7 / −10.8	41 / 42	337 / 118
Maximum audit	0.225 / 0.375	19.4 / 19.9	−7.6 / −9.8	30 / 53	293 / 119
Expectation adjustment time	4.5 / 7.5	20.8 / 19.5	−8.4 / −9.7	35 / 47	253 / 120
Contamination reference	90 / 150	20.9 / 19.6	−8.5 / −10.3	36 / 43	270 / 131
Automation bias	0.1875 / 0.3125	19.7 / 21.7	−10.9 / −8.1	43 / 41	120 / 226
Fuben-eki commitment rate	0.15 / 0.25	19.9 / 21.9	−9.2 / −10.1	33 / 50	208 / 122
Fuben-eki forgetting rate	0.0375 / 0.0625	22.0 / 20.1	−10.1 / −9.3	51 / 35	119 / 214
Quality sensitivity	2.25 / 3.75	20.3 / 21.6	−9.7 / −9.8	42 / 42	146 / 146
Disruption drift magnitude	0.075 / 0.125	20.4 / 21.9	−9.0 / −10.5	43 / 42	140 / 151
Error reference	0.09 / 0.15	21.1 / 20.5	−8.7 / −9.7	37 / 42	258 / 140
Trust shock sensitivity	0.6 / 1	21.4 / 21.2	−10.0 / −9.8	42 / 42	146 / 146
Audit adjustment time	1.5 / 2.5	21.4 / 21.1	−9.8 / −9.9	42 / 42	150 / 143

Trust adjustment speed	0.375 / 0.625	21.1 / 21.4	−9.7 / −10.0	42 / 42	145 / 146	

Baseline values: resilience benefit 21.3 kt CO_2-eq; DiD −9.9 kt CO_2-eq; extra burden 42%; S3 contamination stock 146 t. Parameters are ordered by the largest absolute change in the resilience benefit.

Table S6. Key SD results by scenario (month 1-100, time step 1 month)

Indicator	S0 Conventional, normal	S1 Fuben-eki, normal	S2 Conventional, disrupted	S3 Fuben-eki, disrupted
Peak sorting-error rate, months 30–39	0.118	0.102	0.154	0.131
Sorting-error rate, month 100	0.132	0.122	0.132	0.124
Minimum AI model accuracy	0.29	0.34	0.24	0.32
AI model accuracy, month 100	0.29	0.34	0.29	0.34
Minimum operator skill	0.24	0.29	0.24	0.28
Operator skill, month 100	0.24	0.29	0.24	0.28
Minimum trust in AI	0.29	0.35	0.08	0.10
Maximum fuben-eki state	0.00	0.33	0.00	0.33
Maximum / mean audit intensity	0.124 / 0.063	0.150 / 0.089	0.124 / 0.063	0.150 / 0.090
First month generation > rework capacity	43	91	30	30
Return below rework capacity after disruption (month)	–	–	–	39
Peak contamination stock (t)	616	130	938	161
Contamination stock, month 100 (t)	616	130	938	146
Minimum recycled share	0.21	0.43	0.16	0.40
GHG emissions, month 100 (t CO_2-eq/month)	1,988	1,658	2,068	1,680
Cumulative GHG emissions, month 100 (kt CO_2-eq)	171.8	160.4	182.5	161.3
Cumulative intervention burden (–)	1.27	1.78	1.27	1.81

Resilience benefit (S2 - S3) = 21.3 kt CO_2-eq (11.7%); routine benefit (S0 - S1) = 11.4 kt CO_2-eq; disruption penalty = 10.8 kt CO_2-eq (conventional) and 0.9 kt CO_2-eq (fuben-eki); DiD = −9.9 kt CO_2-eq and −306 t of contamination; extra intervention burden = 42%.

Table S7. Rework-capacity sweep (all other parameter baseline)

Rework capacity (t/month)	S2 stock, month 100 (t)	S3 stock, month 100 (t)	Resilience benefit S2 − S3 (kt CO_2-eq)	Routine benefit S0 − S1 (kt CO_2-eq)	DiD emissions (kt CO_2-eq)	Extra burden (%)	S3 back below capacity after disruption
95	2,796	1,603	9.2	13.3	4.1	41	no
100	2,386	1,254	10.2	14.9	4.7	41	no
105	1,996	909	12.6	16.1	3.5	41	no
110	1,628	564	17.6	16.3	−1.3	41	no
115	1,283	271	22.2	14.9	−7.3	42	month 43
120	938	146	21.3	11.4	−9.9	42	month 39
125	593	124	16.9	7.0	−9.9	43	month 37
130	248	123	8.7	3.1	−5.6	43	no backlog
135	132	123	3.4	2.7	−0.7	43	no backlog
140	132	123	3.1	2.7	−0.4	43	no backlog
145	132	123	2.9	2.7	−0.3	43	no backlog
150	132	123	2.9	2.7	−0.2	43	no backlog
155	132	123	2.9	2.7	−0.2	43	no backlog
160	132	123	2.9	2.7	−0.2	43	no backlog
165	132	123	2.9	2.7	−0.2	43	no backlog

The resilience benefit peaks at 115-120 t/month, where the system operates near its capacity threshold, and falls to about 3 kt CO_2-eq when capacity is ample (≥ 135 t/month), while the extra burden remains at 41-43% across all capacities.

Monte carlo setting

Each sample evaluates all four scenarios with common parameter values (seed 42), so that differences between scenarios are not affected by sampling noise. Spearman rank correlations between the sampled parameters and the resilience benefit (S2 - S3) are weak for every parameter ($|\rho| \leq 0.18$): fuben-eki forgetting rate −0.17; operator forgetting rate 0.14; base concept drift 0.14; maximum audit 0.13; operator learning rate −0.12; ai effect on error −0.11; fuben-eki commitment rate 0.08; error reference −0.07; trust adjustment speed 0.06; audit adjustment time −0.06; trust shock sensitivity −0.05; contamination reference 0.05; human effect on error −0.03; ai learning rate 0.03; quality sensitivity −0.03; expectation adjustment time −0.02; automation bias −0.00; disruption drift magnitude −0.00. Uncertainty in the benefit therefore arises from combinations of parameters that jointly shift the baseline sorting error relative to the rework-capacity threshold rather than from any single parameter, consistent with the capacity sweep in Table S7.

Table S8. Monte carlo results (500 Llatin hypercube samples and 18 parameters varied uniformly bt ±25%

Outcome	Median	5th–95th percentile
Resilience benefit, S2 − S3 (kt CO_2-eq)	14.6	3.6 to 22.2
Resilience benefit (% of S2 emissions)	8.1	2.2 to 12.0
Routine benefit, S0 − S1 (kt CO_2-eq)	9.8	2.8 to 15.6
DiD emissions (kt CO_2-eq)	−5.1	−11.6 to 3.9
DiD contamination stock, month 100 (t)	−211	−359 to −1
Contamination stock S2, month 100 (t)	922	118 to 1,906
Contamination stock S3, month 100 (t)	166	106 to 1,201
Extra intervention burden, S3 vs S2 (%)	38	24 to 55

Probabilities across the 500 samples: P(S3 emissions < S2 emissions) = 1.00; P(routine benefit > 0) = 1.00; P(DiD contamination < 0) = 0.97; P(DiD emissions < 0) = 0.78; P(S3 returns below rework capacity after the disruption) = 0.68.